

Neurolabs Cordova SDK Integration Guide

Technical Integration Specification

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Status: Partner integration reference

Owner: Neurolabs

1. Scope

This guide is for partner hybrid apps integrating 'neurolabs-cordova-sdk'.

2. Changes In v1.1.7

- ? Init and per-session routing use 'taskUUID'.
- ? Native detector/model warmup is enforced in 'init' and guarded in 'openCamera'.
- ? 'openCamera' can return 'MODEL_INIT_FAILED' when native model initialization fails.
- ? 'autoCloseAfterCapture=true' now keeps preview enabled and closes after Save confirmation.
- ? 'cameraClosed' event now includes 'message'.
- ? 'captureQueued' events are deferred while camera is open and flushed after 'cameraClosed'.

3. Install

```
cordova plugin add /path/to/neurolabs-cordova-sdk
```

4. SDK Initialization + Warmup

```
const Neurolabs = cordova.require('neurolabs-cordova-sdk.neurolabs');

await Neurolabs.init({
  apiKey: '<API_KEY>',
  apiUrl: 'https://api.neurolabs.ai/v2',
  taskUUID: '<DEFAULT_TASK_UUID>',
  allowBase64PhotoExport: false
});
```

Warmup is native-side; monitor progress through 'loadingProgress' event.

```
Neurolabs.addListener('loadingProgress', (payload) => {
  console.log('loading', payload);
});
```

5. Queue Management

```
await Neurolabs.setAutoSyncEnabled(true);
await Neurolabs.setWifiOnlyUploadsEnabled(false);
await Neurolabs.setDeletePhotosOnUploadSuccess(true);

await Neurolabs.pauseQueue();
await Neurolabs.resumeQueue();
await Neurolabs.setRetryPolicy({ retryCount: 5 });

const status = await Neurolabs.getQueueStatus();
console.log('queue status', status);

await Neurolabs.flushQueue();
await Neurolabs.retryFailedUploads();
```

6. Open Custom Camera

```
await Neurolabs.openCamera({
  sessionId: crypto.randomUUID(),
  type: 'shelf',
  guidanceMode: 'strict',
  validationPreset: 'ios_parity',

  confidenceThreshold: 0.25,
  iouThreshold: 0.45,
  maxCaptures: 1,

  showAlignmentGuidance: true,
  enableValidation: true,
  liveQualityChecksEnabled: true,
  liveQualityTargetFps: 6,

  // keep custom-camera guidance UI path
  showDetections: false,
  showCapturedRegions: false,

  // optional metadata/cropping
  sendDetectionsMetadata: true,
  enablePreviewCropping: true,

  // now defaults to true if omitted, but explicit is clearer
  autoCloseAfterCapture: true,

  // per-session routing override
  taskUUID: 'bfc85982-b955-4f65-9f32-b6dbed85f364',

  // image size constraints
  maxImageDimension: 1920,
  maxImageWidth: 1920,
  maxImageHeight: 1920,
  imageCompressionQuality: 0.85
});
```

7. Post-Processing + Lifecycle Events

```

Neurolabs.addListener('captureResult', (payload) => {
  // Camera result payload from native side
  console.log('captureResult', payload);
});

Neurolabs.addListener('cameraClosed', ({ sessionId, cancelled, captureCount, message }) => {
  console.log('cameraClosed', sessionId, cancelled, captureCount, message);
});

Neurolabs.addListener('captureQueued', (item) => console.log('queued', item));
Neurolabs.addListener('uploadSucceeded', (item) => console.log('uploaded', item));
Neurolabs.addListener('uploadFailed', (payload) => console.warn('uploadFailed', payload));
Neurolabs.addListener('queueStatusChanged', (status) => console.log('queueStatusChanged', status));

```

8. Notes

- ? 'openCamera' is the custom native camera endpoint.
- ? Use the strict shelf payload above for full shelf guidance checks and auto-close after a validated save.
- ? 'liveQualityChecksEnabled=true' is required for pill/rotation/warning/error guidance behavior.
- ? Keep 'type: 'shelf', 'guidanceMode: 'strict', 'showDetections: false', and 'showCapturedRegions: false' for the custom-camera guidance UI path.
- ? 'guidanceMode: 'guidance' should only be used as a temporary fallback if a partner explicitly wants looser Android behavior than the parity recommendation.
- ? 'autoCloseAfterCapture=true' closes after Save from preview/review flow.
- ? 'captureQueued' may arrive after 'cameraClosed' when capture queueing happens while camera is still active.
- ? Keep base64 transport disabled unless explicitly needed.